YICHEN (ANNA) TU

+1 (984)317-9902 | anna.tu@duke.edu | Durham, NC

EDUCATION

Duke University, Pratt School of Engineering, Durham, NC **Expected May 2027**

Master of Engineering (M.Eng.), Financial Technology

- Relevant Coursework: Programming for Fintech, Quantitative Risk Management, Machine Learning for Fintech

University of St Andrews, St Andrews, Scotland **Sep 2021 – May 2025**

Bachelor of Science (Honours) in Economics and Mathematics

- Relevant Coursework: Causal Inference, Time Series Analysis, Asset Pricing, Mathematical Modeling, Advanced Linear Algebra

University of Virginia (Exchange Student*)*, Charlottesville, VA **Aug 2023 – May 2024**

- Relevant Coursework: Real and Complex Analysis, Econometrics, Stochastic Processes

PROFESSIONAL EXPERIENCE

Fortune SG Fund Management **Shanghai, China**

Global Markets Intern (Investment Department) Jun 2025 – Aug 2025

- Collected, cleaned, and automated large datasets on daily NAV, valuation ratios, and sector exposures using Excel/Python; built templates that reduced manual monitoring time by ~30% and delivered more consistent portfolio tracking for senior analysts
- Calculated risk factors, including volatility, beta sensitivity, and concentration exposures, for multiple ETFs and mutual funds, supported stress-testing scenarios that highlighted downside exposures and informed investment committee meetings discussions
- Performed bottom-up and macro-linked analysis on equity and energy funds (e.g., ARKF, HuaAn Oil & Gas LOF), connecting with OPEC+ output and market news, and synthesized findings into fund presentations and investor roadshow materials

Hwabao Securities **Shanghai, China**

Mutual Fund Research Intern (Research and Innovation Department) Jun 2024 – Aug 2024

- Processed stock weights and factor exposure datasets across 9 indices (~1,000 data points per stock) using Python, built the foundation for a multi-factor risk model, supported portfolio exposure analysis and visualized key insights
- Validated mutual fund and index datasets, applied Pandas to compute daily PnL, historical VaR, and exposure breakdowns; designed Excel dashboards with charts, pivot tables, and slicers to track portfolio risk metrics and present findings to senior
- Analyzed ~50,000 raw data points using multivariate regression to identify operational risk factors that significantly influenced process time—such as returned data and stage-specific approval differences—and assisted in streamlining process management
- Acted as liaison among Investment, IT, and Risk teams for better communications and cross-team collaborations

PROJECTS AND LEADERSHIP

Fintech 545 Quant Risk Management Class Project (Python)

- **Market Risk:** Conducted comprehensive market risk analysis including Value at Risk (VaR) and Expected Shortfall (ES) assessments across multiple portfolios, informing risk management strategies in line with regulatory and business needs.
- **Financial Product Modeling:** Priced various financial products using DCF (bonds), FRAs (swaps), and Monte Carlo/Black Scholes (options), enhancing portfolio accuracy and decision-making.

Standardization and IPW for Causal Inference - Honours Dissertation (R Studio)

- Estimated public health risk of smoking on overweight incidence using BRFSS 2023 data (~400,000 obs); implemented both standardization and inverse probability weighting (IPW) methods in R to assess robustness across modeling assumptions
- Found ATE estimates ranging from −5.5% (IPW) to −9.3% (standardization); highlighted robustness in risk quantification

Corporate Innovation Research (Python)

- Integrated Clarivate (1TB patents, assignees, inventors) and Crunchbase (employment) datasets by Stata(reclink2) and Python
- Improved probabilistic record linkage accuracy by 5–17%, enhancing reliability of worker–company matching
- Delivered cleaner, more accurate datasets that enabled faculty research on corporate innovation and long-term productivity trends

STEP Diving into Data 2024 (R Studio)

- Directed a 7-member team in a university data challenge analyzing 3,000+ ADHD case records and presented to mentors
- Conducted EDA and preprocessing in R (data merging, missing-value handling, numeric parsing, categorical encoding)
- Found measurable associations between soil contamination, heavy metal exposure, and ADHD incidence, demonstrating how data-driven methods can uncover health risk factors

ADDITIONAL SKILLS

Technical: Python (pandas, NumPy, scikit-learn), R (tidyverse, ggplot2), SQL, MATLAB, C++, LaTeX